

ST PAUL ISLAND, AK, USA

WMO: 703080

Lat: 57.155N

Lon: 170.222W

Elev: 35

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	1.5	5.7	-4.0	4.4	2.0	0.3	5.6	6.4	40.7	22.8	37.5	25.8	16.2	10	0.965

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	55.4	52.4	53.9	51.5	52.5	50.6	53.3	54.6	52.1	53.2	51.1	52.1	13.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.4	58.6	53.7	51.3	56.2	52.6	50.3	54.0	51.6	22.2	54.7	21.5	53.2	20.9	52.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.9	34.0	30.8	0.1	59.7	7.2	4.2	-5.0	62.7	-9.2	65.1	-13.3	67.4	-18.5	70.5		
(5)			-0.6	54.8	7.2	2.6	-5.8	56.6	-10.0	58.1	-14.0	59.6	-19.2	61.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	25.4	25.9	25.1	29.9	36.4	42.8	47.5	49.3	46.1	39.4	34.1	29.2	
	DBStd	10.46	8.70	9.19	9.01	6.25	4.27	3.32	2.97	2.41	3.36	4.38	5.06	6.86	
	HDD50	5153	764	675	771	603	421	217	89	43	120	329	477	644	
	HDD65	10592	1229	1095	1236	1053	886	666	543	488	566	794	927	1109	
	CDD50	36	0	0	0	0	0	1	11	20	4	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	15.5	18.4	18.4	17.0	16.2	13.6	12.1	11.2	12.2	14.5	16.3	18.1	17.8	
Precipitation	PrecAvg	23.3	1.7	1.3	1.2	1.2	1.1	1.3	1.9	2.9	2.8	3.0	2.9	2.2	
	PrecMax	35.3	5.0	3.3	3.3	3.2	2.8	3.1	4.3	5.5	5.7	6.2	7.7	4.6	
	PrecMin	9.8	0.3	0.0	0.1	0.1	0.1	0.0	0.4	0.2	0.6	0.9	0.9	0.6	
	PrecStd	4.7	0.9	0.7	0.8	0.7	0.6	0.7	0.9	1.1	1.2	1.2	1.2	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	39.4	39.5	42.5	49.4	55.5	58.5	57.6	55.1	50.2	45.2	42.1	
		MCWB	37.9	38.2	37.5	39.5	43.8	49.5	54.3	54.2	52.3	48.2	43.2	40.6	
	2%	DB	38.0	37.9	37.9	40.2	45.7	51.4	55.3	55.5	53.1	48.7	43.9	40.2	
		MCWB	36.8	36.6	36.3	38.1	42.2	47.4	52.4	52.9	50.7	46.7	42.4	38.8	
	5%	DB	36.9	36.7	36.4	38.8	43.7	49.4	53.4	54.1	51.8	47.1	42.7	39.0	
		MCWB	35.8	35.4	35.0	37.1	41.1	46.2	51.2	52.0	49.7	45.0	41.3	37.6	
	10%	DB	35.7	35.7	35.0	37.3	42.1	47.8	51.8	53.0	50.6	45.5	41.4	37.8	
		MCWB	34.5	34.4	33.7	35.7	40.2	45.2	50.1	51.2	48.7	43.5	39.8	36.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.8	38.4	40.4	44.8	50.8	55.1	55.1	53.4	49.3	44.3	41.2
			MCDB	38.7	39.1	39.0	41.5	47.8	54.3	57.4	56.9	54.4	49.6	44.7	41.8
2%		WB	37.1	37.1	36.7	38.8	43.0	48.2	53.1	53.6	51.4	47.6	43.1	39.2	
		MCDB	37.7	37.5	37.4	39.8	45.0	50.7	54.6	54.7	52.2	48.6	43.6	39.8	
5%		WB	36.0	35.8	35.2	37.3	41.6	46.7	51.5	52.6	50.5	45.9	41.7	37.9	
		MCDB	36.7	36.4	36.1	38.4	43.2	48.8	52.7	53.5	51.3	46.9	42.5	38.7	
10%		WB	34.7	34.6	33.8	36.0	40.4	45.8	50.4	51.5	49.6	44.2	40.1	36.7	
		MCDB	35.5	35.3	34.6	37.0	41.9	47.7	51.5	52.4	50.4	45.4	41.2	37.7	
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	7.6	6.9	7.2	7.4	6.2	5.9	6.7	6.9	6.3	7.0	
		MCDBR	5.1	5.7	6.8	7.0	9.5	10.1	8.7	7.4	7.2	6.3	6.3	6.2	
		MCWBR	5.3	5.9	6.5	6.0	6.9	6.9	6.2	5.6	5.8	5.8	6.8	6.4	
	5% WB	MCDBR	4.8	5.7	6.6	6.4	8.8	8.9	7.7	6.5	5.6	5.4	6.1	6.2	
		MCWBR	5.2	6.2	6.5	5.8	6.6	6.4	6.0	5.3	5.1	5.6	6.8	6.6	
Clear-Sky Solar Irradiance	taub	0.297	0.336	0.332	0.400	0.408	0.403	0.392	0.388	0.370	0.338	0.325	0.271		
	taud	2.211	2.205	2.287	2.097	2.201	2.306	2.402	2.412	2.359	2.319	2.145	2.320		
	Ebn at Noon	179	220	260	255	262	266	266	259	244	218	158	164		
	Edh at Noon	16	26	30	43	41	38	34	31	29	23	17	11		
All-Sky Solar Radiation	RadAvg	125	321	746	1121	1313	1358	1171	967	756	442	161	81		
	RadStd	15	36	122	128	96	97	88	98	59	28	18	11		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page